

Construction Replay Transcript

EDI 837 NCCI MUE — Design · 2026-08-13 · what you'll hear in documents/construction-replay.mp4

What this interface does

Show that PilotFish can apply CMS Medically Unlikely Edits (MUE) as a business-rule layer on X12 837P claims: look up each service line's CPT/HCPCS against an external MUE table and kick out lines whose units exceed the max. There is no stock PilotFish NCCI/MUE module — this is the same catalog + XSLT pattern already used in Med Rec (MUE_EDITS.CPT / MAX_VALUE_PER_LINE), pointed at 837 instead of HL7 DFT.

Who it's for

- Sources: Directory-drop 837P (input/inbound/*.edi)
- Destinations: output/pass/ (within MUE limit), output/kickout/ (units exceeded)
- Demo vs production: Demo only — synthetic CMS-shaped table (not the official quarterly NCCI files). Production would load CMS MUE CSVs into SQL (or a file lookup) on the CMS publish cycle.
- Sibling demo: edi-837-ncci-ptp (procedure-to-procedure pairs). Med Rec does not implement PTP.
- Anyone watching the construction video or reading this transcript

Systems involved

- Sources: Directory-drop 837P (input/inbound/*.edi)
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Documents we also produce

EDI837_NCCI_MUE_Capability_Brief.pdf · ready

Short overview of what the interface does — shareable without reading route XML.

EDI837_NCCI_MUE_Test_Plan.pdf · ready

Scenarios that prove the interface works end-to-end.

Module documentation pack (10 deep-dive PDFs) · ready

PilotFish product PDFs for the modules used in this build.

EDI837_NCCI_MUE_V2_Route_Diagrams.pdf · ready

V2 route diagrams for visual review of module topology.

construction-replay.mp4 + this transcript · ready

Narrated walkthrough of building the routes, then a live Demo-tab test when inject is available.

Narration

Spoken lines from the construction video (setup, build-replay, live test, closing).

Setup

Welcome

Welcome to the PilotFish demo: EDI 837 NCCI MUE. Show that PilotFish can apply CMS Medically Unlikely Edits (MUE) as a business-rule layer on X12 837P claims: look up each service line's CPT/HCPCS against an external

MUE table and kick out lines whose units exceed. PilotFish supports the whole flow out of the box.

Pipeline overview

Before we wire anything, here's the flow end to end. Show that PilotFish can apply CMS Medically Unlikely Edits (MUE) as a business-rule layer on X12 837P claims: look up each service line's CPT/HCPCS against an external MUE table and kick out lines whose units exceed.

External systems

And here's what sits around the routes. SQL Server, PilotFish eiPlatform, and Demo Web UI. All of those are spun up in docker images for this demo.

1 - Poll And Score MUE

First route

Show that PilotFish can apply CMS Medically Unlikely Edits (MUE) as a business-rule layer on X12 837P claims: look up each service line's CPT/HCPCS against an external MUE table and kick out lines whose units exceed. We'll build that in 2 routes. Starting with a blank canvas on the first route.

Poll EDI 837P Files

Here's a directory listener — it watches a local folder. That folder is `/opt/pilotfish/input/inbound`. We're only taking `.edi`, `.txt`, `.837` files. It checks about every 5 seconds. After pickup we move the file aside so the folder stays clean.

Fork - XPath

This XPath fork splits the file so each transaction set becomes its own message.

Extract Claim Identity

XPath evaluation copies a few fields off the XML onto the transaction so later steps can use them.

Debug Write Forked Transaction

This writes a debug copy of the XML so we can inspect what the route saw. That lands under `/opt/pilotfish/output/debug`.

Extract Claim Line XML

Now for the mapping — we're using the stock XSLT processor with `extract-claim-line.xslt`. The output is a claim line with claim control number, procedure code, and units.

Read Claim Line Fields

XPath evaluation copies a few fields off the XML onto the transaction so later steps can use them.

Lookup MUE Catalog

This is a SQL lookup — we query `Edi837NccMue` over JDBC using a key from the transaction.

Extract MUE Catalog Fields

XPath evaluation copies a few fields off the XML onto the transaction so later steps can use them.

Build MUE Decision XML

Now for the mapping — we're using the stock XSLT processor with build-mue-decision.xslt. Values from earlier steps come in as parameters. The output is an MUE decision with claim control number, procedure code, units, max units, and catalog found. Then it sets pass or kickout.

Debug Write MUE Decision

This writes a debug copy of the XML so we can inspect what the route saw. That lands under /opt/pilotfish/output/debug.

Pretty-Print Staged Decision

This processor pretty-prints the XML — line breaks and indent — so the file on disk is readable.

Stage MUE Decision

Then we write the file out to /opt/pilotfish/output/staged-decisions.

2 - Route MUE Decisions

Next route

Next route — blank canvas again, then the modules one at a time.

Staged-folder listener

Here's a directory listener — it watches a local folder. That folder is /opt/pilotfish/output/staged-decisions. We're only taking .xml files. It checks about every 3 seconds.

Extract Decision Attributes

XPath evaluation copies the match bucket off the decision XML so the router can fan out.

Conditional Router

Then a conditional router — it uses XPath to pick which path this transaction takes.

Pretty-Print Pass XML

This processor pretty-prints the XML — line breaks and indent — so the file on disk is readable.

Pretty-Print Kickout XML

This processor pretty-prints the XML — line breaks and indent — so the file on disk is readable.

Write Pass

Then we write the file out to /opt/pilotfish/output/pass.

Write Kickout

Then we write the file out to /opt/pilotfish/output/kickout.

Live test

After construction, the video switches to the Demo tab and runs a live inject when the UI supports it.

Prove it works

Routes are built. Let's switch to the Demo tab and prove it works.

Submit a sample

I'll submit this sample from the Demo tab — same as an operator would.

Waiting for the routes

Give the routes a moment to pick up the file and write the output.

MUE catalog

There it is — output on the results panel.

Submit another sample

I'll send it one more time so we can watch it land again.

Waiting for the routes

Same path — the second result should show up here.

MUE catalog

And there's the second result.

Closing

Demo complete

That's the demo — show that PilotFish can apply CMS Medically Unlikely Edits (MUE) as a business-rule layer on X12 837P claims: look up each service line's CPT/HCPCS against an. Thanks for choosing PilotFish.